

Problem 10.13

The force of interest depends upon time, with

$$\delta(t) = \frac{1}{10(1+t)}, \quad 0 \leq t \leq 4.$$

Find the accumulated amount after four years for an initial investment of \$1,000 at time zero.

Solution

For a time-varying force of interest, the accumulation factor from time 0 to time 4 is

$$a(4) = \exp \left(\int_0^4 \delta(t) dt \right).$$

Substitute the given force of interest:

$$a(4) = \exp \left(\int_0^4 \frac{1}{10(1+t)} dt \right).$$

Evaluate the integral:

$$\int_0^4 \frac{1}{10(1+t)} dt = \frac{\ln(1+t)}{10} \Big|_0^4.$$

Thus,

$$\int_0^4 \frac{1}{10(1+t)} dt = \frac{\ln 5 - \ln 1}{10} = \frac{\ln 5}{10} = 0.1609437912.$$

Therefore, the accumulated amount is

$$A(4) = 1000e^{\ln(5)/10}.$$

Equivalently,

$$A(4) = 1000(5)^{1/10}.$$

Hence,

$$\boxed{A(4) = 1174.62}.$$